Golam Ishtiak

isht0001@e.ntu.edu.sg — LinkedIn — ishtiakk.github.io — +65 9086 6474

Education

Nanyang Technological University (NTU), Singapore *Aug 2024 – May 2028*
Bachelor of Science in Physics, Second Major in Data Analytics, Minor in Mathematics **GPA: 4.83/5.00**
Nanyang Scholarship Recipient: One of NTU's most prestigious undergraduate merit scholarships.
Relevant Coursework: Quantum Mechanics II, Statistical Mechanics, General Relativity, Physics of Classical & Quantum Information, Algorithms & Computing, Probability. *In progress:* Graduate Quantum Information, Condensed Matter Physics I, Geometric Methods in Mathematical Physics.

McGill University, Montreal, Canada *Jan 2027 – Apr 2027*
Incoming Semester Exchange (Department of Physics)

Awards & Achievements

International & Asian Physics Olympiads (representing Bangladesh): Bronze Medal at IPhO 2023 (ranked 119th of 394) and 2022 (ranked 140th of 368); Honorable Mention at APhO 2022 (ranked 62nd of 224).
PLANCKS Competition (Singapore Round): Runner-up in 2026.
UK Integration Bee (Singapore Round): Top 10 Finalist in 2025 and 2026.

Research Experience

Hardware-Aware EMG Inference Using Filamentary Memristor Synapses *Sep 2026 – Present*
Supervisor: Prof. Lew Wen Siang, NTU

- Performed photolithography, etching, and direct-write laser lithography on silicon wafers in a cleanroom environment, working toward filamentary memristors as synaptic weight-storage elements.
- Planned work: electrically characterize fabricated devices (conductance states, programming variability, state stability) and integrate measured non-idealities into a neural network simulation to evaluate inference robustness for edge-AI EMG processing.

Thermalization in 1D Ultracold Gases *Aug 2025 – Present*
Supervisor: Prof. Nelly Ng Huei Ying, NTU (URECA Programme)

- Analyzed 1D Bose gas dynamics via the Lieb-Liniger model, applying third-order perturbation theory and Wick decomposition to truncate the interaction hierarchy and derive phase-phase correlation equations.
- Simulated non-equilibrium relaxation using Truncated Wigner Approximation (TWA) trajectories, applying convex optimization (SDP) to enforce the Heisenberg Uncertainty Principle on the symplectic covariance matrix.
- Quantified bipartite continuous-variable entanglement via the symplectic eigenvalues of the partially transposed covariance matrix, tracking the sudden death of logarithmic negativity during dephasing (forthcoming in **NTU URECA Proceedings 2025–26**); currently characterizing quantum information leakage into non-Gaussian modes.

Review of Relativistic Quantum Mechanics *May 2026 – Aug 2026*
Supervisor: Dr. Leek Meng Lee, NTU (Odyssey Research Programme)

- Conducted a theoretical review of quantum mechanical formalism and relativistic matter fields (Klein-Gordon and Dirac equations); presented a poster at the Odyssey Research Programme (July 2026).

Professional & Leadership Experience

Mercor Intelligence *Remote*
Project Lead — Frontier AI Physics Evaluation *May 2026 – Oct 2026*

- Led a 25-member team of PhDs and Olympiad medalists to test frontier AI models by directing the creation of an adversarial physics benchmark; shipped 100 complex problems that exposed reasoning failure modes across leading LLMs (yielding model accuracy scores from 0% to 90%).
- Managed project milestones, enforced quality control across deliverables, and served as the primary liaison between subject-matter experts and the technical engineering team.

Physics Team Co-Lead (Olympiad Section) & Evaluator *Jul 2025 – Jan 2026*

- Co-led a 15-member physics subgroup within a 50+ expert division developing training data for a frontier GenAI model. Designed and delivered ∼500 Olympiad-level problems and solutions for reinforcement learning workflows.

Student Ambassador & Peer Tutor, NTU SPMS *Feb 2026 – Present*

- Guide incoming scholars through admission to NTU, and provide peer tutoring for undergraduate SPMS courses.

Physics Olympiad Trainer & Academic Instructor *Jan 2023 – Present*

- Trained Bangladesh National Physics Olympiad camps, coaching the 2024 and 2025 IPhO teams and 2023 IJSO team (contributing to 8 International Medals and 5 Honorable Mentions combined).
- Produced online STEM outreach lectures and open-source notes on Olympiad-level physics and mathematics.

Technical Skills & Additional Information

Programming & Tools: Python (NumPy, Pandas, Matplotlib, CVXPY), C, R, Mathematica, LaTeX.
Laboratory: Cleanroom fabrication (photolithography, etching).
Languages: English (Fluent), Bengali (Native) — **Interests:** Competitive Programming, Cycling, Hiking.